Threat Hunting in Finance Sector with MITRE ATT&CK

By

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Project Overview

This project focuses on proactive threat hunting within the Finance industry, leveraging the MITRE ATT&CK framework to identify and analyse Advanced Persistent Threat (APT) groups targeting the sector.

Introduction

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Aims and Approach

The central objective is to employ threat-hunting methodologies to address the unique security challenges facing financial organisations. By adopting the MITRE ATT&CK framework, the project systematically examines the tactics, techniques, and procedures (TTPs) used by APT groups that specifically target the finance sector.

Through this structured approach, the project aims to enhance understanding of the threat landscape, pinpointing the methods adversaries use to compromise financial systems. This enables a more informed and strategic response to emerging threats.

Scope of Analysis

The investigation is centred on mapping identified APT groups to relevant TTPs, as catalogued in the MITRE ATT&CK Navigator. By comparing and visualising these attack patterns, the project seeks to uncover commonalities among the threat actors, thereby supporting the development of targeted defensive measures within the financial industry.

Objectives

- Understand the MITRE ATT&CK framework and its practical application in threat hunting scenarios.
- Investigate advanced persistent threats (APTs) that are targeting the finance sector by utilising SOCRadar Labs.
- Link the identified APT groups to corresponding tactics, techniques, and procedures (TTPs) in the MITRE ATT&CK Navigator.

- Conduct a comparative analysis to identify and emphasize shared attack patterns.

Tools and Resources

SOCRadar Labs

SOCRadar Labs is utilised to obtain threat intelligence that is specific to the Finance sector. This resource enables the identification and investigation of Advanced Persistent Threats (APT) groups that pose risks to financial organisations.

MITRE ATT&CK Navigator

The MITRE ATT&CK Navigator is employed to map the tactics, techniques, and procedures (TTPs) associated with identified APT groups. This tool facilitates a visual and comparative analysis of attack patterns relevant to financial institutions.

MITRE ATT&CK Framework

The MITRE ATT&CK Framework provides a structured taxonomy of adversary behaviours. By referencing this framework, the project systematically categorises the tactics and techniques used by threat actors targeting the finance sector.

Project Steps

1. Understanding the MITRE ATT&CK Framework

The initial phase of the project involved a comprehensive study of the MITRE ATT&CK framework's structure. This framework serves as a foundation for analysing adversary behaviours and mapping attack methodologies relevant to the Finance sector.

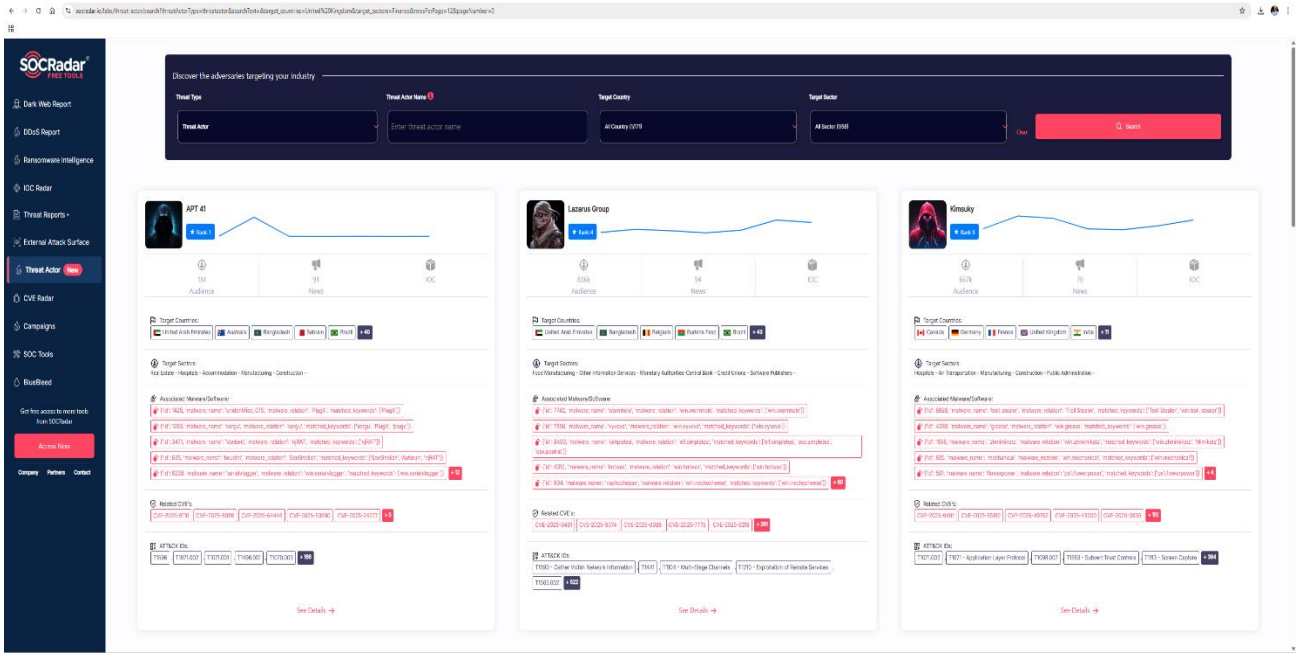
- **Tactics:** These represent the underlying objectives or motivations behind an attack, such as Initial Access, Persistence, and Defence Evasion.
- **Techniques:** Techniques detail the specific methods employed by threat actors to achieve their objectives—for example, phishing or credential dumping.

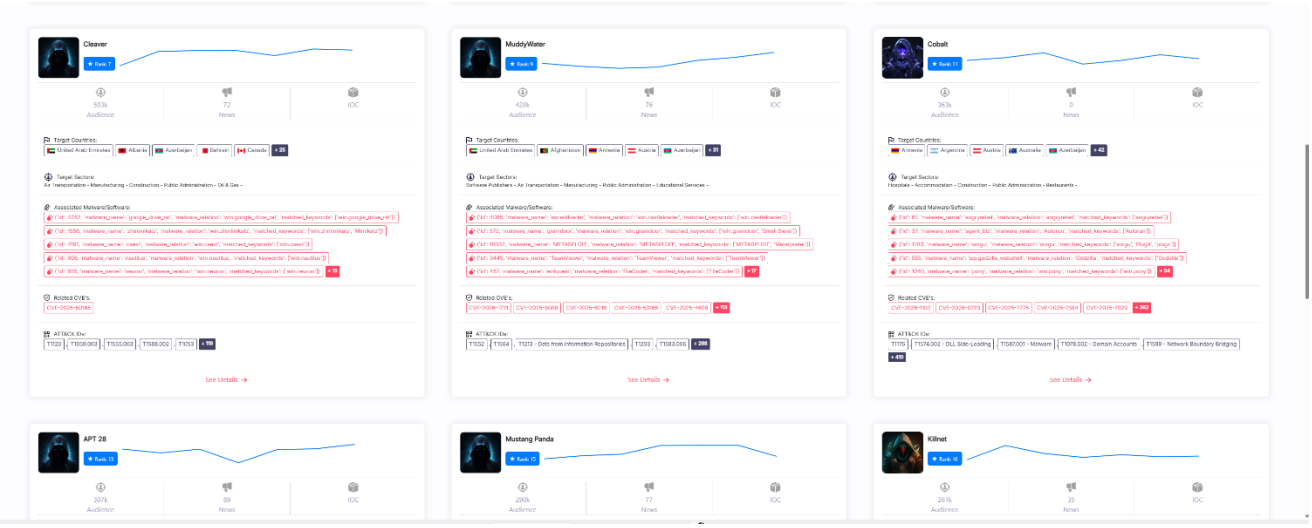
- **Procedures:** Procedures are the real-world implementations of these techniques, illustrating how adversaries apply them in practical scenarios.

2. Researching Advanced Persistent Threats (APTs) Specific to the Finance Sector

In this phase, the project focused on identifying Advanced Persistent Threat (APT) groups that are known to target the finance sector. By examining threat intelligence and utilising relevant research tools, the aim was to pinpoint the adversaries whose tactics, techniques, and procedures (TTPs) pose a significant risk to financial organisations. This research stage involved methodically analysing the available data to ensure that the APT groups selected were pertinent to the sector under study.

• Used SOCRadar Labs to identify APT groups targeting





List of the ATPs found using the SOCradar

1. APT 41 also known as Amoeba, BARIUM, BRONZE ATLAS, Blackfly, Brass Typhoon, Double Dragon, Earth Baku, G0044, G0096, Grayfly, HOODOO, LEAD, Leopard Typhoon, Red Kelpie, TA415, TG -2633, WICKED PANDA, WICKED SPIDER, Winnti
2. Lazarus group also known APT 38, APT-C-26, Appleworm, BeagleBoyz, Black Artemis, Bluenoroff
3. Kimsky also known as MoneyLibra
4. Cobalt also known as Cobalt group.

3. Identified Advanced Persistent Threats (APTs) Targeting the Finance Sector

The following APT groups were identified through SOCradar Labs as being active in targeting financial organisations. Each group is known by multiple aliases, reflecting their presence across various threat intelligence sources:

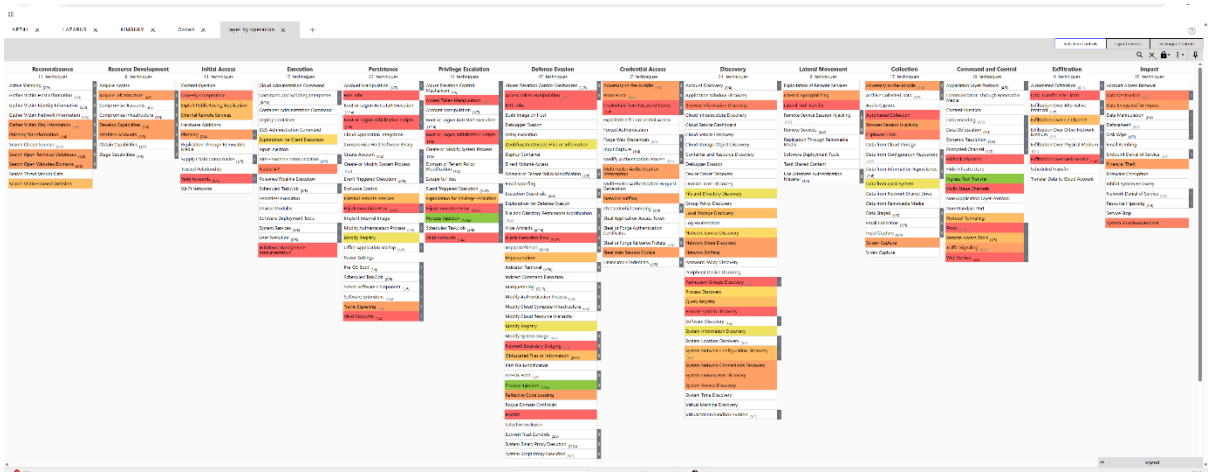
- **APT 41**: This group is also referred to as Amoeba, BARIUM, BRONZE ATLAS, Blackfly, Brass Typhoon, Double Dragon, Earth Baku, G0044, G0096, Grayfly, HOODOO, LEAD, Leopard Typhoon, Red Kelpie, TA415, TG2633, WICKED PANDA, WICKED SPIDER, and Winnti. Their numerous aliases highlight the breadth of their operations and the complexity of tracking their activities.

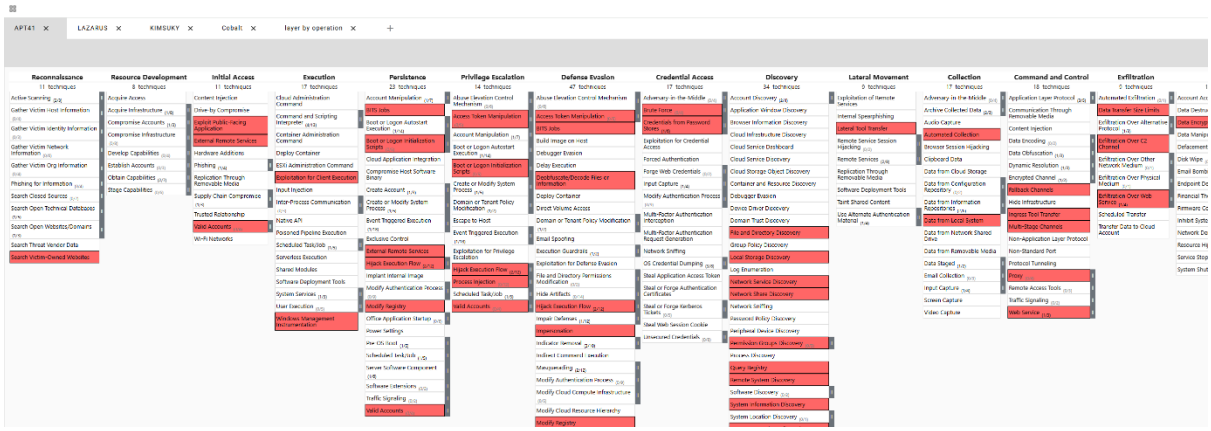
- **Lazarus Group**: Also known as APT 38, APTC26, Appleworm, BeagleBoyz, Black Artemis, and Bluenoroff. These aliases reflect the group's various campaigns and its reputation for targeting financial institutions with sophisticated cyber operations.
- **Kimsuky**: This group is alternatively referred to as MoneyLibra. The alias is used in several threat intelligence reports to describe its financial sector-focused tactics.
- **Cobalt**: Known also G0080, Cobalt spider, Cobalt gang

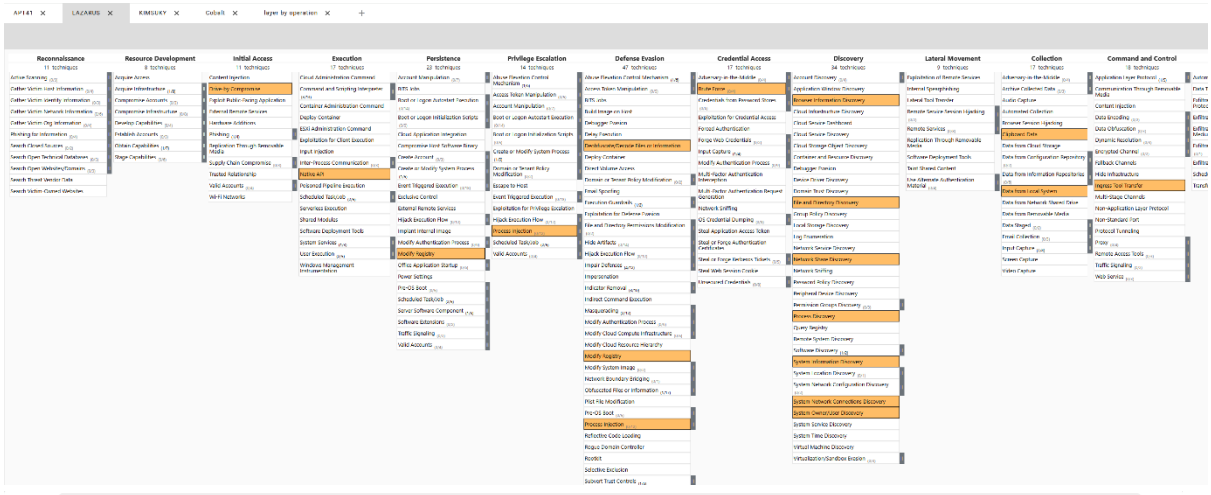
These groups represent significant threats to the finance sector, utilising a range of tactics, techniques, and procedures tailored to exploit vulnerabilities within financial organisations.

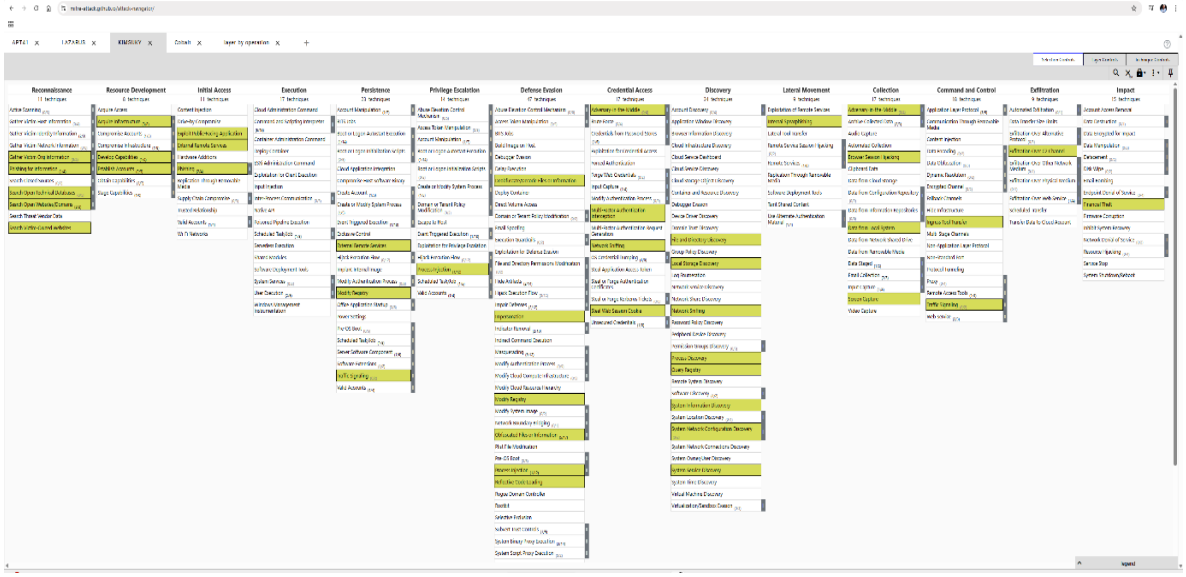
4. Map APTs to TTPs using MITRE Navigator

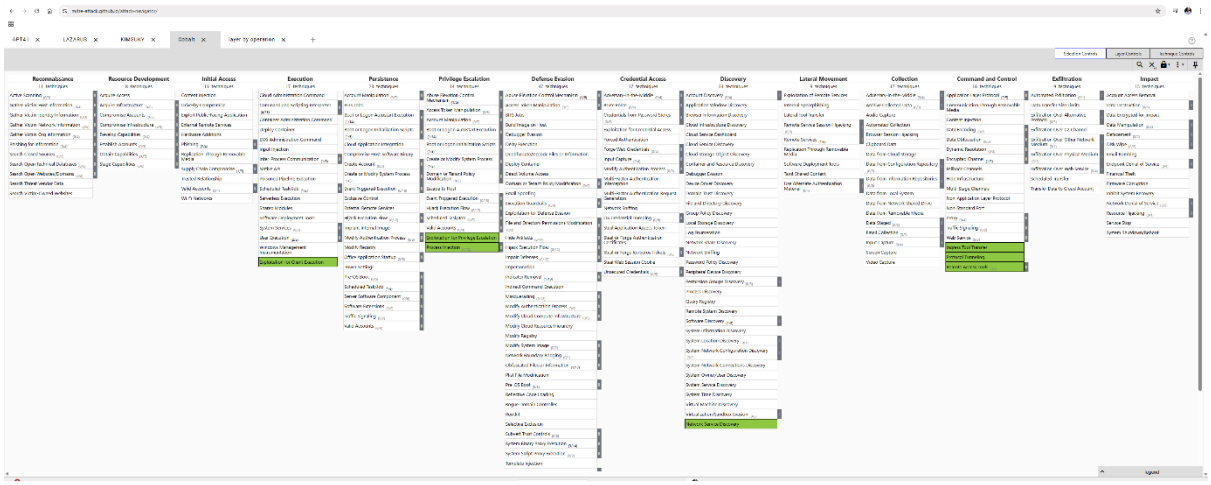
Utilising the MITRE ATT&CK Navigator, individual layers were generated for each of the four identified APT groups with the United Kingdom specified as their operational location. This approach enabled a clear visual representation of the tactics, techniques, and procedures (TTPs) associated with each group. The creation of separate layers facilitated comparative analysis, making it possible to observe the specific methods used by these threat actors and to identify areas of overlap or similarity in their attack patterns targeting the finance sector.




APT41
LAZARUS
KIMSUKY
Cobalt
layer by operation
Reconnaissance
Resource Development
Initial Access
Execution
Persistence
Privilege Escalation
Defense Evasion
Credential Access
Discovery
Lateral Movement
Collection
Command and Control
Exfiltration
Impact







5. Findings

Analysis revealed several key patterns and trends:

- APT groups targeting finance frequently employ tactics such as Initial Access (spear phishing, watering hole attacks), Credential Access (credential dumping, brute force), and Lateral Movement (remote desktop protocol exploitation).
- There is significant overlap in TTPs among groups, with spear phishing and credential dumping emerging as the most prevalent techniques.
- Cobalt and Lazarus Group often exploit transactional systems and financial platforms, while APT 41 demonstrates broad targeting of both infrastructure and user accounts.
- Visualisation highlighted clusters of techniques concentrated around persistence and privilege escalation, suggesting adversaries prioritise maintaining access and expanding control once initial compromise is achieved.

These insights underscore the importance of prioritising defence against phishing and credential theft, strengthening monitoring of lateral movement, and ensuring robust authentication protocols. The recurrence of certain TTPs suggests that financial organisations should focus on layered security controls and regular threat intelligence updates to adapt to evolving adversary behaviour.

Conclusion and Recommendations

The project demonstrates that mapping APTs to TTPs using the MITRE ATT&CK framework provides valuable visibility into the threat landscape affecting the finance sector. Financial organisations should leverage these insights to inform security policy, prioritise detection and response capabilities, and invest in targeted employee training (e.g., phishing awareness). Regular updates to threat intelligence and continuous review of attack patterns are essential to maintaining resilience against advanced threats. Policy makers are encouraged to facilitate sector-wide sharing of threat intelligence and promote adoption of frameworks like MITRE ATT&CK for structured, collaborative defence.